

Scalability & Future Plan

Date: 29 june 2026

Team ID: (xxxx)

Project Name: Human Development Index (HDI) Prediction Using Machine Learning

Maximum Marks: 1 Mark

Scalability & Future Plan

The **Human Development Index (HDI) Prediction Using Machine Learning** project is designed with a scalable architecture using **Python, Flask, Scikit-learn, Pandas, NumPy, HTML, CSS, and JavaScript**. The current system provides accurate HDI predictions based on key development indicators. In the future, the application can be enhanced to support larger datasets, more users, cloud deployment, and additional analytical features to improve performance and usability.

Current System Limitations:-

S.No	Limitation	Impact	Priority to Address
1	Application is hosted on a local Flask development server.	Cannot support many concurrent users or internet access.	High
2	Uses a static CSV dataset for prediction.	Data may become outdated over time.	High
3	No user authentication or rolebased access control.	Limited security for future enterprise use.	Medium

Scalability Plan:-

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports local users only.	Deploy on cloud platforms such as Render, Railway, or AWS to support multiple users simultaneously.
2	Data Storage	Stores data in CSV files.	Migrate to a relational database such as MySQL or PostgreSQL for better scalability and data management.
3	Performance	Single machine prediction.	Optimize the machine learning model, implement caching, and use a production WSGI server (Gunicorn/Waitress).
4	Security	Basic input validation only.	Add user authentication, HTTPS, encrypted data transfer, and role-based access control.

Future Roadmap:-

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the Flask application to a cloud hosting platform.	1–2 Months	Makes the application accessible online for users from anywhere.
Phase 2	Integrate real-time HDI datasets from trusted public sources.	3–4 Months	Improves prediction accuracy using updated information.
Phase 3	Add interactive dashboards, graphs, and country comparison features.	4–6 Months	Enhances data visualization and decision-making capabilities.

Phase 4	Implement user authentication, report generation, and mobile responsiveness.	6–8 Months	Improves security, usability, and overall user experience.
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